Automated Construction Specification Review with Named Entity Recognition Using Natural Language Processing

Seonghyeon Moon[1]; Gitaek Lee[2]; Seokho Chi, M.ASCE[3]; and Hyunchul Oh[4]

Abstract: When bidding on construction projects, contractors need to understand the specifications properly to manage project risks. However, specifications are mainly analyzed based on human cognitive abilities, which can take considerable time and can lead to errors due to misunderstanding. While efforts have been made to automate this process, that the existing academic efforts to automate the process have limitations. To develop an automated specification reviewing model applicable to various kinds of specifications, the authors propose information extraction frameworks consisting of five categories. In addition, a named entity recognition (NER) model was developed based on bidirectional long short-term memory architecture to extract information from text data automatically. The data set for model development comprised 56 construction specifications, which included a total of 4,659 sentences labeled according to five categories of information. Word2Vec was utilized to aconvert labeled text data to the form of numeric vectors to be input into the NER model. The NER model successfully assigned every word in the testing data to an appropriate category with a satisfactory performance of 0.919 precision and 0.914 recall. These results contribute to the automation of the construction specification review process. Although this research focused on road construction projects, the proposed information extraction framework can be applied to other types of construction projects.
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Introduction

A construction specification is a document that specifies the qualitative requirements for performing work during a construction project, many of which are related to technical construction issues. The document commonly consists of specifications, drawings, contracting requirements, and procurement requirements. Because the construction specifications written by the client are legally binding contracts, contractors must follow the requirements in the specifications precisely during the construction of a project. If the contractor fails to satisfy the specifications, problems can occur, such as the increased risk of accidents, rework of documents, wasted resources, construction delay, poor performance, conflicts among stakeholders, and lawsuits (Zhang and El-Gohary 2017; Zhong et al. 2012). Eventually, the project can incur wasted time and human resources as well as increased costs.

A clear and objective understanding of construction specifications during the bidding stages is particularly important for contractors to manage project risks. Because construction specifications contain the provisions that have to be satisfied within the project, contractors should review the document contents thoroughly and analyze each clause to ensure it is appropriate and realistic considering the site conditions (Lam et al. 2007). However, contractors generally experience difficulties when reviewing specific details manually because of the large volume of information (often more than several thousand pages), inconsistent hierarchies of clauses, and variety of provisions in terms of formats and meanings. Because of these properties, the review process is often time-consuming and costly, requiring significant human effort, and the results are vulnerable to subjectivity and review errors. To overcome these problems, many researchers have attempted to address the issue of how to automatically analyze construction documents by applying natural language processing (NLP) and text-mining techniques (Malsane et al. 2015; Moon et al. 2019; Salama and El-Gohary 2016).

However, the previous approaches included several critical limitations in terms of applicability. In particular, they manually developed information extraction rules, analyzed only a few types of construction activities, and were restricted to specific clauses in a specification. Because construction specifications come in different formats and include different types of provisions, it is necessary to develop a new model that can review a variety of construction specifications. To develop a robust model to automate the process of reviewing construction specifications, the authors proposed an information extraction framework consisting of user-needed information and types of construction specifications, and developed a machine learning–based information extraction model to provide the information automatically. First, several recent studies were reviewed to clarify the gap between manual and automated processes for reviewing construction specifications. Second, an automated framework using machine learning–based named entity recognition (NER) based on bidirectional long short-term memory (Bi-LSTM) architecture was proposed. Third, five information types were

[1]Postdoctoral Research Associate, Dept. of Civil and Environmental Engineering, Seoul National Univ., 1 Gwanak-Ro, Gwanak-Gu, Seoul 08826, Republic of Korea; Senior Researcher, Institute of Construction and Environmental Engineering, Seoul National Univ., 1 Gwanak-Ro, Gwanak-Gu, Seoul 08826, Republic of Korea. Email: blank54@snu.ac.kr

[2]Graduate Student, Dept. of Civil and Environmental Engineering, Seoul National Univ., 1 Gwanak-Ro, Gwanak-Gu, Seoul 08826, Republic of Korea. Email: lgt0427@snu.ac.kr

[3]Professor, Dept. of Civil and Environment Engineering, Seoul National Univ., 1 Gwanak-Ro, Gwanak-Ku, Seoul 08826, Republic of Korea; Adjunct Professor, Institute of Construction and Environmental Engineering, 1 Gwanak-Ro, Gwanak-Ku, Seoul 08826, Republic of Korea (corresponding author). Email: shchi@snu.ac.kr

[4]Senior Researcher, Smart Construction Team, Daewoo E&C, 20 Suil-Ro 123beon-Gil, Jangan-Gu, Suwon 16297, Republic of Korea. Email: hyunchul.oh@daewooenc.com

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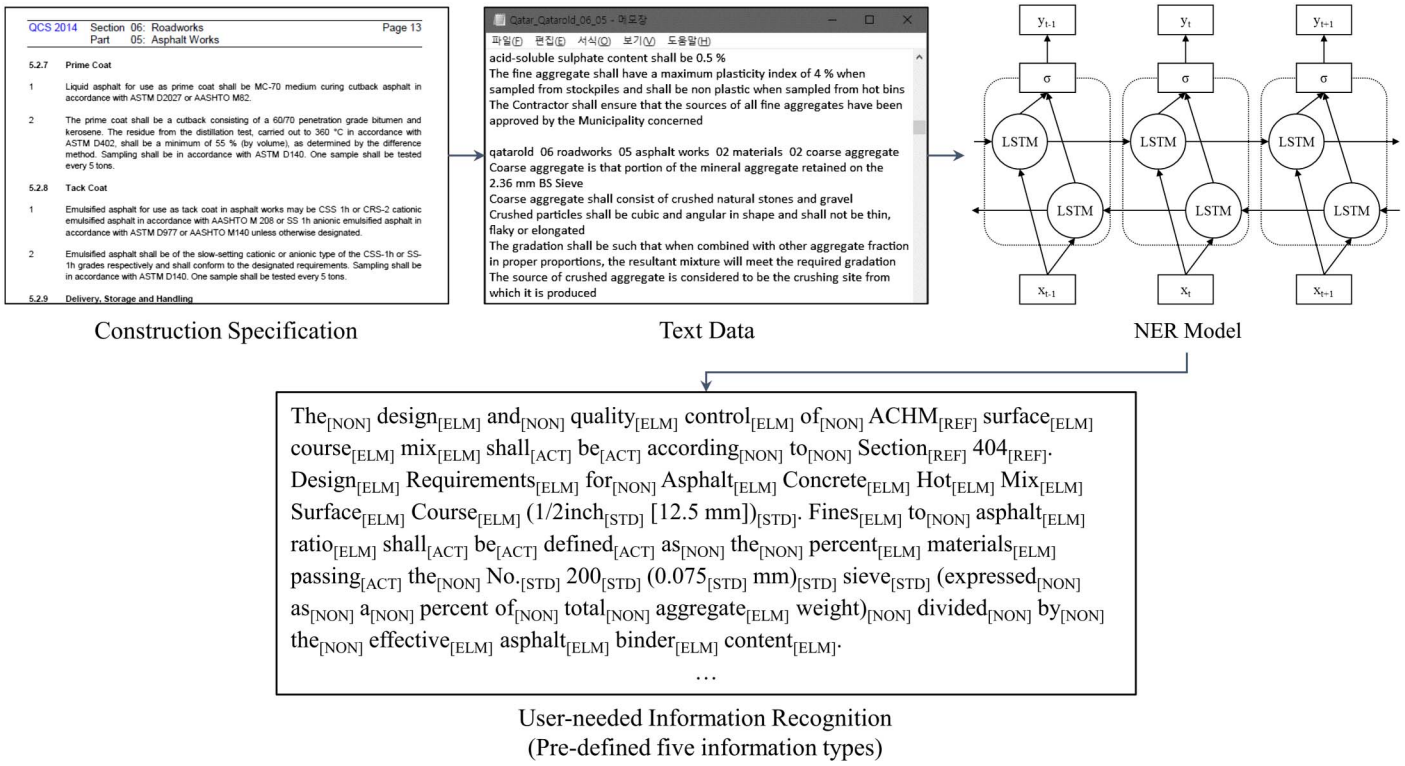


Fig. 1. Conceptual research framework.

defined to understand the contents of construction specifications, including people and organizations in charge, activities required, construction and installation items, quality standards and criteria, and relevant references. Finally, an NER model was developed to automatically extract the user-needed information (i.e., the five predefined information types) from text data. Fig. 1 shows the conceptual framework of the research, including the input and output elements of each research step.

The results of this research will contribute to the construction body of knowledge. The NER model successfully extracted the user-needed information from the specifications. The construction specifications could be understood automatically based on these information types regardless of volume, format, and organizational structures of clauses. Furthermore, the specifications could be reviewed quickly without any omission of information, and the results retained consistency. In addition, the research will also contribute to the construction industry by providing necessary information for contract risk management during the bidding and planning stages, quality management during the construction stage, and conflict management after project completion. Although the research scope was limited to the specifications for road construction projects, the five information types suggested by the research may also be useful when automatically reviewing other types of construction specifications.

Literature Review

Automated Analysis of Construction Documents

Many efforts have been made using NLP and text mining to automatically provide user-needed information from construction documents (Kim and Chi 2019; Solihin and Eastman 2015; Zhang and El-Gohary 2014). Initial attempts included developing a document classification system for construction and facility management projects (Caldas and Soibelman 2003; Craig and Sommerville 2006; Kerrigan and Law 2005). These attempts included categorizing the documents by construction item (such as materials, space, and physical boundaries) for which data were already available in a structured format. In other words, the results were restricted to the items on which they focused; thus, it was impossible to acquire data for items that were not listed.

Automatic compliance checking is a process of automatically assessing the compliance of construction documents with applicable laws and regulations. Salama and El-Gohary (2013), one of the pioneers, proposed an approach for analyzing the provisions of laws and regulations in construction while understanding semantic information. Afterward, the results were enhanced by advanced applications of information transformation (Zhang and El-Gohary 2015), rule-based information extraction (Zhang and El-Gohary 2016), and ontology and deontology (Zhang and El-Gohary 2017). Eventually, a fully automated system for automatic compliance checking was developed and tested with the International Building Code (Zhang and El-Gohary 2017). However, these previous studies showed three fundamental improvement opportunities. First, the information extraction rules were built manually. In other words, researchers with adequate domain knowledge read text data, extracted common features, determined the patterns from the features, and built the rules by listing the patterns within the provisions for analysis. The extracted features, for example, included syntactic patterns (e.g., "subject," "subject restriction," "quantitative relation," and "quantity value") and conformance information (e.g., "obligation," "permission," "prohibition," and "forbidden"). Several rules were formulated by mixing these patterns, and the rules were used to extract information from the documents (Salama and El-Gohary 2013; Zhang and El-Gohary 2016). Such approaches showed good performance when the data size was limited to a small volume; however, the longer and the more complicated a document, the more rules were required. Second, only a

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few types of construction elements were analyzed; that is, these approaches only focused on very specific and restricted problems. Third, the results were restricted to a small amount of data, such as one or two chapters from one or two documents; they were not appropriate for use with other documents or even with other chapters in the same document because the patterns only suited the original data.

To overcome the limitations of the previous studies, some researchers developed information extraction models using machine learning approaches (Kim and Chi 2019; Moon et al. 2019). Although the manually built rules have the advantage of being comprehensive and accurate for the analyzed data, they require significant labor in development and the rules cannot be applied to other data sets (Beach et al. 2015; Lee et al. 2019). However, the machine learning models can automatically extract rules by training the input and output and determining useful features from the data. Even if the machine learning–based rules are difficult to understand, the rules strengthen automation solutions and enhance expandability (Al Qady and Kandil 2015). As the amount of training data increases, the model learns to extract features and information important to users.

Semantic Named Entity Recognition

NER is a subfield of the machine learning–based information extraction methodologies and labels each word with predefined, informative categories, such as name, location, and object (McCallum and Li 2003; Sang and De Meulder 2003). These categories, called named entities, are defined by researchers in order to comprehend the text data based on the categories. For example, in the construction industry, the NER model was developed to extract construction information, such as regulatory items (Zhang and El-Gohary 2016), bridge defect information (Liu and El-Gohary 2017), and accident information (Kim and Chi 2019).

NER is divided into two approaches based on what information is used within the text data: syntactic information or semantic information. First, NER that uses syntactic information is known to perform well with small and clean data sets because the syntactic expressions in a sentence that are needed to determine the category of each target word can be easily extracted from such data sets (Newman et al. 2006). For instance, words associated with the name category would have an initial capital, words associated with the location category would appear right after a preposition (such as in, on, or to), and words associated with the object category would, in many cases, be nouns. This approach shows satisfactory accuracy if the text data are in a clean and standard format. Zhang and El-Gohary (2016) defined which object to extract from text sentences (e.g., subject, subject restriction, quantitative relation, quantity value) and developed extraction rules based on phrase structure grammar. Liu and El-Gohary (2017) proposed an NER model to extract bridge damage information, such as bridge elements, deficiency types and causes, and maintenance activities. Kim and Chi (2019) calculated the conditional probability of each word's role in a sentence and extracted the accident keywords, including hazard object, hazard position, work process, and accident result.

Although such syntactic NER models provided promising results in the extraction of informative keywords, they all required ontologies (i.e., the relationship between text words) to build extraction rules. On the other hand, NER using semantic information is well known for its robustness and expandability compared to the syntactic approach. The model automatically identifies the usage patterns of each word within the text and acquires the semantic information from the patterns based on a machine learning algorithm (Cucerzan and Yarowsky 1999; Ratinov and Roth 2009).

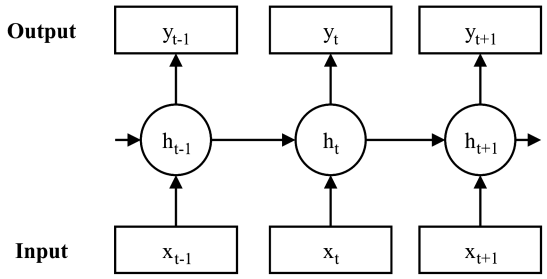


Fig. 2. RNN model architecture.

For instance, if a corpus (i.e., a group of documents) consists of sentences such as "The Contractor should prepare the equipment" and "The document should be submitted by the Engineer," the semantic NER model would consider usage patterns and classify "Contractor" and "Engineer" in the same category, even though the syntactic role of each word is different. This characteristic enables the semantic NER approach to be less limited with respect to the document quality or variety of writing styles than the syntactic approach. In this research, a semantic NER model was developed to deal with the large volume and variety of text data in terms of meaning and formats in construction specifications.

Recurrent Neural Network for NER

The recurrent neural network (RNN) is one of the deep neural networks, where the networks are connected in a series structure (Nallapati et al. 2016). Due to these serialized networks, the RNN model can use the sequential information of the input data, and thus outperform other machine learning models in the analysis of serialized data, such as NER. Fig. 2 illustrates the basic architecture of an RNN, where t, x, h, and y indicate the sequence step, layer input, layer output, and output class, respectively.

Unlike structured data, text data (i.e., the sequence of words) have context information with words that have close relationships. Hence, text data sometimes have different meanings, even with the same spelling. For instance, the term "mixed" can be used as a verb (i.e., an action, such as "should be mixed in"); however, in another case, the same term can be used as an adjective (i.e., modifying a noun, such as "the mixed design"). Conventional machine learning models that do not have a serialized structure cannot differentiate these examples [e.g., see the term "mixed" in Fig. 3(a)]. However, the RNN model can address this problem by treating the input data as a sequence of words; that is, the RNN model can classify the category of the term with higher accuracy by considering the context information of the text data [e.g., see the term "mixed" in Fig. 3(b)].

Although the RNN model shows strength in analyzing text data, it also has a critical limitation called the vanishing gradient problem. That is, the longer the network serializes, the smaller the gradient becomes, which is vital for delivering past information forward and obtaining updates by back-propagation; hence, the learning ability of the model decreases severely. The long short-term memory (LSTM) concept has addressed such limitations (Hochreiter and Schmidhuber 1997). LSTM networks contain a unique structure (i.e., forget gate and input gate) within the hidden layer of the RNN (Fig. 4). The forget gate, f_t, plays a role in forgetting the past information, whereas the input gate, i_t, plays a role in remembering the current information. These structures enable the model to conserve essential information for a longer distance by forgetting unnecessary signals while reinforcing necessary ones (Wu et al. 2016). In this research, Bi-LSTM architecture that considers the sequential information in both forward and backward

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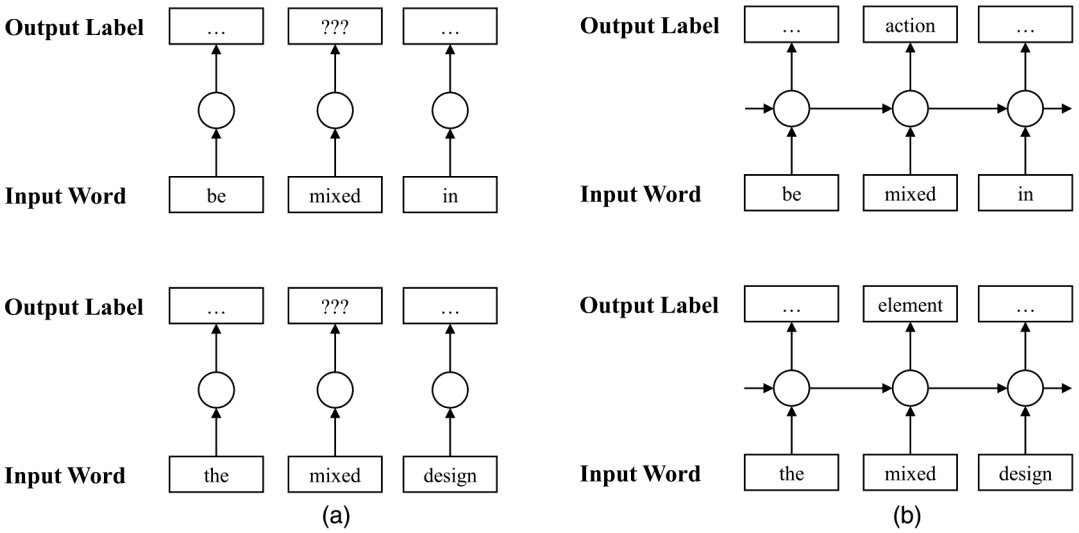


Fig. 3. Example of word classification result: (a) conventional classification model; and (b) RNN model.

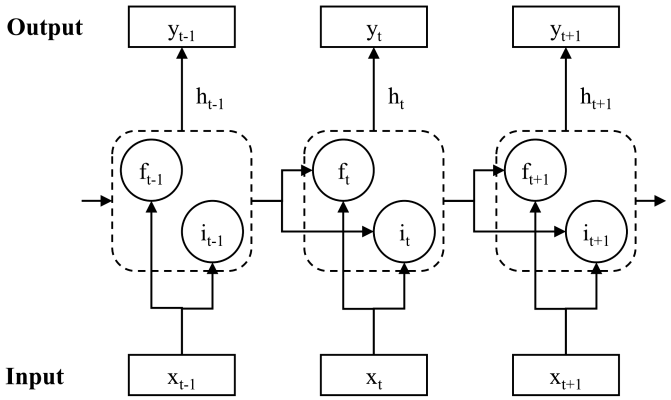


Fig. 4. Simple architecture of LSTM.

ways was used to develop the NER model. A detailed description of the model is given in the following sections.

NER Model Development for Construction Specification Reviews

Model Development Process

This research proposed an automatic information extraction model based on NER. Fig. 5 illustrates the overall procedure of model development. First, the construction specifications were collected and converted into a text file format for analysis. After text tokenization as a preprocess, each word token was mapped to an identical numeric vector by Word2Vec, which converted text data into a computer-readable vector format. Next, the researchers manually labeled every word token to a predefined category to train the model. Finally, the Bi-LSTM model was developed, which eventually predicted the category of each word.

Data Collection

A total of 56 construction specifications were collected for analysis: two were practical specifications used in construction projects performed by Korean contractors (road construction projects in

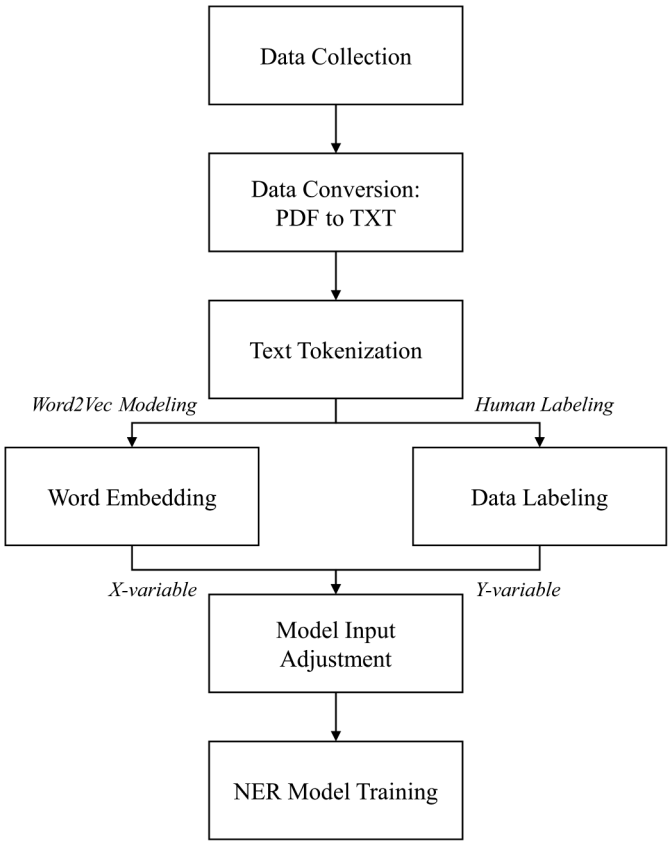


Fig. 5. Model development procedure.

Qatar during 2010 and 2014); and the remaining 54 specifications were national (or regional) standards, which could be fundamentally applied to any road construction project in the selected country (or region). Because most developed countries have their own standard specifications well organized, the research team preferentially collected the latest specifications from the United States, the United Kingdom, Canada, and Australia. The standard specifications were collected from national or government websites. Table 1 lists the collected data.

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Table 1. Data collection

Country	Regulatory authority	Last edited
Qatar	—	2010
Qatar	—	2014
United States	Alabama	2018
United States	Alaska	2017
United States	Arizona	2008
United States	Arkansas	2014
United States	California	2015
United States	Colorado	2017
United States	Connecticut	2018
United States	Delaware	2016
United States	Florida	2018
United States	Georgia	2013
United States	Idaho	2018
United States	Indiana	2018
United States	Kentucky	2012
United States	Louisiana	2016
United States	Maine	2014
United States	Maryland	2008
United States	Massachusetts	1995
United States	Michigan	2012
United States	Minnesota	2018
United States	Mississippi	2017
United States	Missouri	2017
United States	Montana	2014
United States	Nevada	2014
United States	New Hampshire	2016
United States	New Jersey	2007
United States	New Mexico	2014
United States	New York	2018
United States	North Carolina	2018
United States	North Dakota	2014
United States	Ohio	2018
United States	Oklahoma	2009
United States	Oregon	2018
United States	Pennsylvania	2016
United States	Rhode Island	2013
United States	South Dakota	2015
United States	Tennessee	2015
United States	Texas	2014
United States	Utah	2017
United States	Vermont	2018
United States	Virginia	2016
United States	Washington	2018
United States	West Virginia	2017
United States	Wyoming	2010
United Kingdom	England	2018
Canada	Alberta	2013
Canada	British Columbia	2016
Canada	New Brunswick	2015
Canada	Newfoundland and Labrador	2013
Canada	Nova Scotia	2014
Canada	Ontario	2018
Canada	Prince Edward Island	2019
Australia	Northern Territory	2017
Australia	Tasmania	2017
Australia	Western Australia	2018

Text Tokenization

Preprocessing of text data is commonly composed of tokenization, stop word removal, and stemming (or lemmatization) (Manning et al. 2008; Zou et al. 2017). Tokenization parses every sentence into a sequence of words in order to utilize each word for the analysis. For example, the sentence "The specification includes provisions about a technical requirement" would be tokenized into eight words: "the," "specification," "includes," "provisions," "about," "a," "technical," and "requirement." In this research, only the tokenization technique was applied in order to input the original content into the NER model and evaluate its prediction performance even with less-cleaned data sets. In this approach, the research considered words that did not contribute to understanding textual information but that were useful for gleaning context information, such as prepositions, pronouns, and adverbs.

Word Embedding

Text data are written in the form of natural language, which computers cannot analyze. Because the computer requires numeric data input, an embedding process is essential in text analysis. The embedding process involves mapping the data from the natural language space to the numeric vector space (Manning et al. 2008). After embedding, every text data set has its own location in the vector space, which means the text is represented in a numeric vector that the computer can utilize.

Term frequency (TF) is the simplest and most intuitive text-embedding technique. This approach counts the frequency of each word in each document and considers it as a document vector (Manning et al. 2008). For example, two sentences, "The Contractor should prepare" and "The Engineer should submit," would be mapped to $[1_{\text{the}}, 1_{\text{Contractor}}, 1_{\text{should}}, 1_{\text{prepare}}, 0_{\text{Engineer}}, 0_{\text{submit}}]$ and $[1_{\text{the}}, 0_{\text{Contractor}}, 1_{\text{should}}, 0_{\text{prepare}}, 1_{\text{Engineer}}, 1_{\text{submit}}]$. A more enhanced text-embedding technique is term frequency–inverse document frequency (TF-IDF), which normalizes the prevalent terms that are spread throughout almost every document (e.g., a, an, and the) and confers less importance to those words (Joulin et al. 2016; Zhou and El-Gohary 2016). For example, the two preceding sentences would be mapped to $[0.5_{\text{the}}, 1_{\text{Contractor}}, 0.5_{\text{should}}, 1_{\text{prepare}}, 0_{\text{Engineer}}, 0_{\text{submit}}]$ and $[0.5_{\text{the}}, 0_{\text{Contractor}}, 0.5_{\text{should}}, 0_{\text{prepare}}, 1_{\text{Engineer}}, 1_{\text{submit}}]$, respectively, so that the meaningful terms (i.e., "Contractor" and "Engineer") can be used to characterize each document more effectively. While these frequency-based approaches can conduct text embedding efficiently (i.e., by simply counting the frequency of each term), they have a critical limitation in that they do not take into account the context information (i.e., the order or sequence of words).

Recently, machine learning–based embedding techniques have gained popularity in addressing the limitations of the frequency-based approaches. Among the state-of-the-art techniques, the most widely used text-embedding technique is Word2Vec (Mikolov et al. 2013). Word2Vec learns the distributed representation of words within every sentence (i.e., usage patterns) and maps similarly used words to a close vector space. For instance, if there are two sentences, "the Contractor should prepare" and "the Engineer should prepare," Word2Vec locates "Contractor" and "Engineer" in a close vector space. There are two Word2Vec architectures: continuous bag of words (CBOW) and skip-gram (Le and Mikolov 2014; Mikolov et al. 2000). CBOW tries to predict the current word from the surrounding words by controlling a specific size of the window (i.e., the number of surrounding words) [Fig. 6(a)]. For example, with the preceding sample sentence, "the" and "should" are the words surrounding "Contractor" with a window size of 1. The CBOW model finds the most robust projection matrix that receives n surrounding word vectors and predicts the target word vector [i.e., Word_t in Fig. 6(a)] by adjusting the window size and the projection matrix repeatedly. On the other hand, the skip-gram model tries to predict the surrounding words based on the current word [Fig. 6(b)]. According to the developer, CBOW is faster, while the skip-gram infers text better (Google Code Archive 2013). This current study developed the Word2Vec model using the skip-gram

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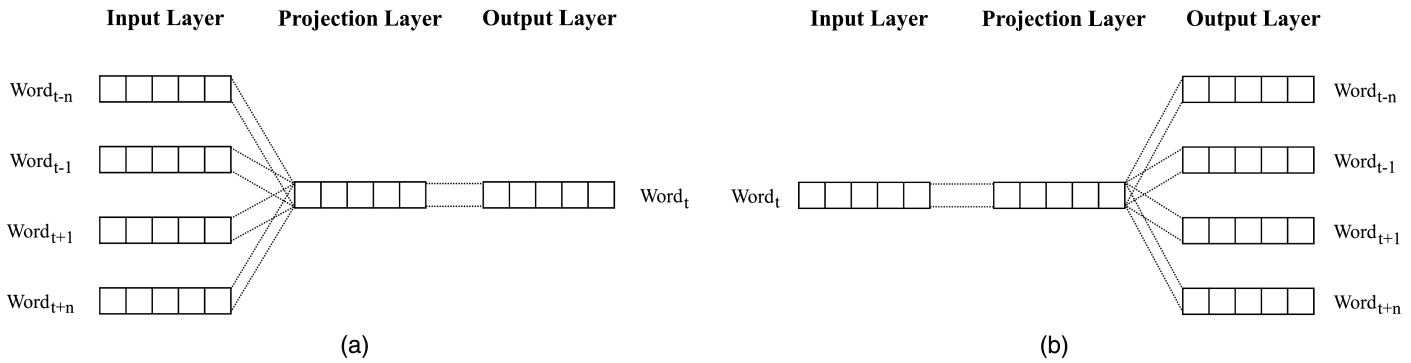


Fig. 6. Word2Vec architecture: (a) CBOW; and (b) skip-gram.

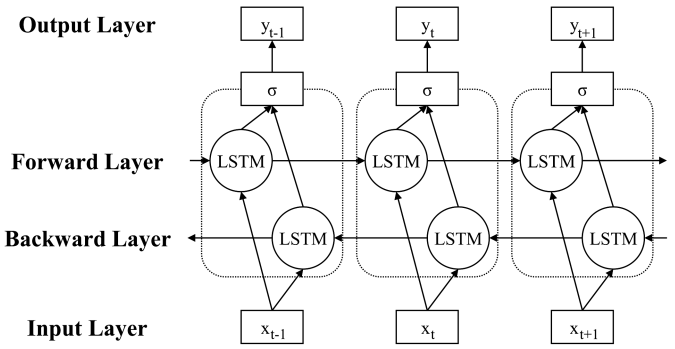


Fig. 7. Architecture of Bi-LSTM.

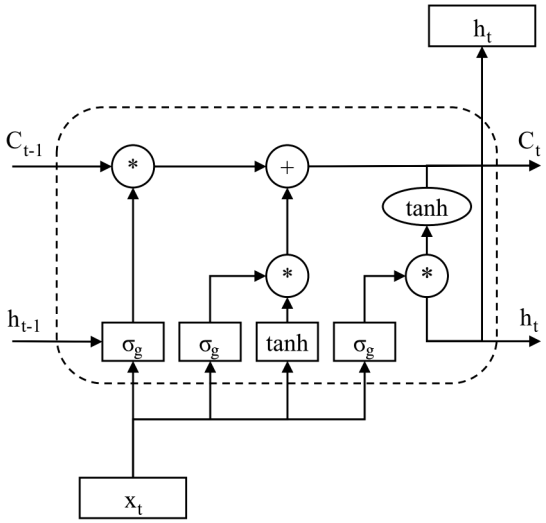


Fig. 8. Detailed structure of the LSTM unit.

architecture for text embedding because the embedding quality would significantly affect the performance of the NER model.

Bi-LSTM Architecture

The Bi-LSTM architecture was applied in developing the NER model after conducting a thorough literature review on RNN, as described in the RNN section of the literature review. Bi-LSTM contains two LSTM layers (i.e., forward and backward), and the output sequences of the two layers were combined using the concatenating function (σ) (Fig. 7) (Cui et al. 2018).

For a detailed structure of the LSTM at time t, the LSTM unit contains the input vector x_t and the layer output h_t, where f_t, i_t, and o_t indicate the forget gate, the input gate, and the output gate, respectively (Fig. 8). The cell state considers the cell input state, $\widetilde{C}_t$, the cell output state, C_t, and the previous cell output state, C_{t-1}. The weight matrices are Ws and Us, and bs are the bias vectors. Finally, σ_g indicates the activation function [Eqs. (1)–(6)]

$$f_t = \sigma_g(W_f \cdot x_t + U_f h_{t-1} + b_f) \tag{1}$$

$$i_t = \sigma_g(W_f \cdot x_t + U_i h_{t-1} + b_i) \tag{2}$$

$$o_t = \sigma_g(W_o \cdot x_t + U_o h_{t-1} + b_o) \tag{3}$$

$$\widetilde{C}_t = \tanh(W_C \cdot x_t + U_C \cdot h_{t-1} + b_C) \tag{4}$$

$$C_t = f_t * C_{t-1} + i_t * \widetilde{C}_t \tag{5}$$

$$h_t = o_t * \tanh(C_t) \tag{6}$$

Information Needs for Construction Specifications: Case Study in Qatar

The specification for a road construction project in Qatar in the Middle East contained 4,790 pages, 29 major categories (i.e., divisions such as Concrete and Road Works), and 285 subcategories (i.e., sections such as Concrete Road Pavements, Concrete Plants, and Curing). At the section level, the categories were not mutually exclusive; for instance, the Concrete Road Pavements section was located under the Road Works division rather than the Concrete division. At the clause level, most sentences were too long to be understood (e.g., more than four to five lines in length without a period), and the same construction elements or objects (e.g., coarse aggregate, curing temperature, and tack coat) sometimes referred to different quality standards. Thus, contractors may experience difficulties when reviewing, understanding, and analyzing the specification's contents. This might also lead to errors, important issues being overlooked, or misinterpretation of the text, likely resulting in unexpected conditions during actual construction in the field.

The authors collaborated with 10 contractors with an average of 15 years of experience to understand the information needs for risk management and identify four types of questions that should be answered by the specification:

1. Who is responsible? For instance, who is responsible for the maintenance of equipment: the engineer or contractor?

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Table 2. Identification of information types

Questions for risk management	Information	User-needed information types
1. Who was responsible?	Who	(1) People and organizations in charge
2. What should be done when?	What	(3) Construction and installation items
	When	(4) Quality standards and criteria
3. How should the construction be?	How	(2) Activities required
	How	(4) Quality standards and criteria
4. Which reference should be followed?	Which reference	(5) Relevant references

Table 3. Examples of NER categories

Information type	NER category	Examples
Organization	ORG	Contractor, engineer, designer
Action	ACT	Must submit, have to approve, shall test
Element	ELM	Formula, certification, design value
Standard	STD	1 month, a week, 38 mm
Reference	REF	AASHTO, ASTM, BS EN ISO
None	NON	And, to, for

2. What should be done by when? For example, the pavement surface should be cleaned at least 3 days before pouring the cement.
3. How should the construction be carried out? For example, the edges should be slopped at gradients not exceeding 10%.
4. Which quality standards should be followed? For example, the sampling of aggregates shall be done in accordance with AASHTO T2 or T248, ASTM C50, or equivalent as applicable.

If these types of questions are not clearly answered, or such contents do not satisfy the local standards during the planning phase before construction, contractors can face risks during the actual construction phase, potentially causing rework, cost overruns, or project delays.

The contractors finally determined that the following information types are crucial to answer the identified questions: (1) people and organizations in charge, (2) activities required, (3) construction and installation items, (4) quality standards and criteria, and (5) relevant references (Table 2). The crucial content required for risk management can be understood based on the determined information types. The authors confirmed that every informative keyword from the specification could be assigned to one of those five categories.

The organization category explained subjects, participants, and stakeholders, such as contractor, engineer, and designer. The Action (ACT) category covered information about how the standard should be met, so it usually included words that corresponded to verb phrases, such as "must submit," "have to approve," and "shall test." The Element (ELM) entity referred to the construction element that was utilized at the site, and it was usually mentioned as the object of the standards. The ELM category included formulas, the names of materials and equipment, and documents. The Standard (STD) category referred to the actual criteria that the organizations must follow or that the construction elements should satisfy. The STD category was usually related to numerical values, such as 1 month, a week, and 38 mm. The Reference (REF) category comprised every document, specification, and code referenced in the text, including AASHTO, ASTM, and the British and European standards (BS EN ISO). To consider the remaining words that were not assigned to one of the first five meaningful categories, the last category, None (NON), was added to the named entities. Various meaningless words, such as "and," "to," and "for," were assigned to the NON category. Table 3 provides examples of each NER category.

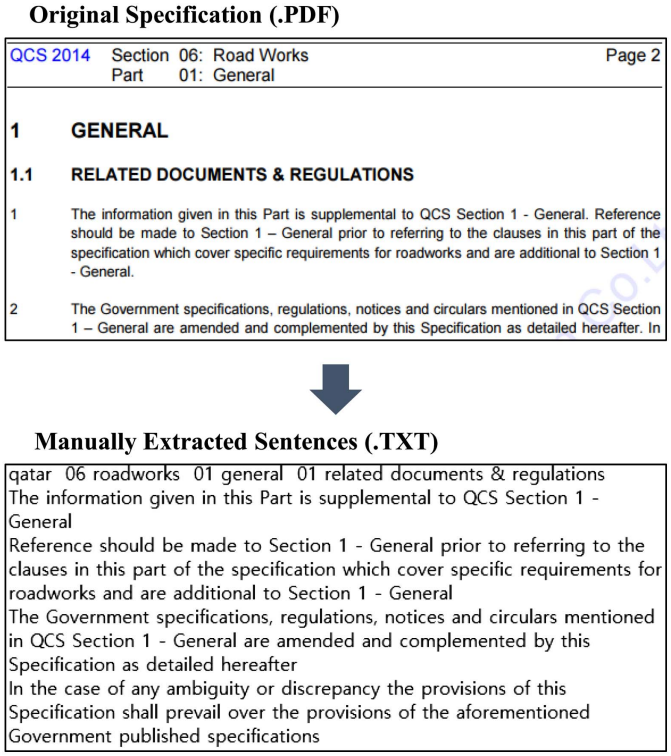


Fig. 9. Data conversion (PDF to TXT).

Experimental Setup

Data Conversion: PDF to TXT

The collected specifications were in PDF format; as such, the computer could not modify or analyze the contents. Therefore, every PDF file was converted to TXT format, which allowed the text to be modified. Initially, the authors utilized open-source software products for automatic file conversion: the pdftotext function in XpdfReader version 4.02 developed by Glyph & Cog; an online platform PDFtoText; and Python library pdftotext. However, all the results were inadequate for further text analysis because the converted TXT files included too many incorrect text recognitions, space errors, unnecessary punctuation marks, and meaningless symbols. Because the quality of the data had an immediate effect on the analysis results, the conversion processes were performed manually; that is, the drag-copy-paste approach was conducted to extract every sentence from the PDF file one at a time (Fig. 9).

Data Labeling

The information types determined from the case study in Qatar referred to five named entities—ORG, ACT, ELM, STD, and REF—which were used as informative word categories in the NER model. Six construction practitioners were involved in manually assigning word labels to the 4,659 construction specification sentences according to the defined categories to be used for training

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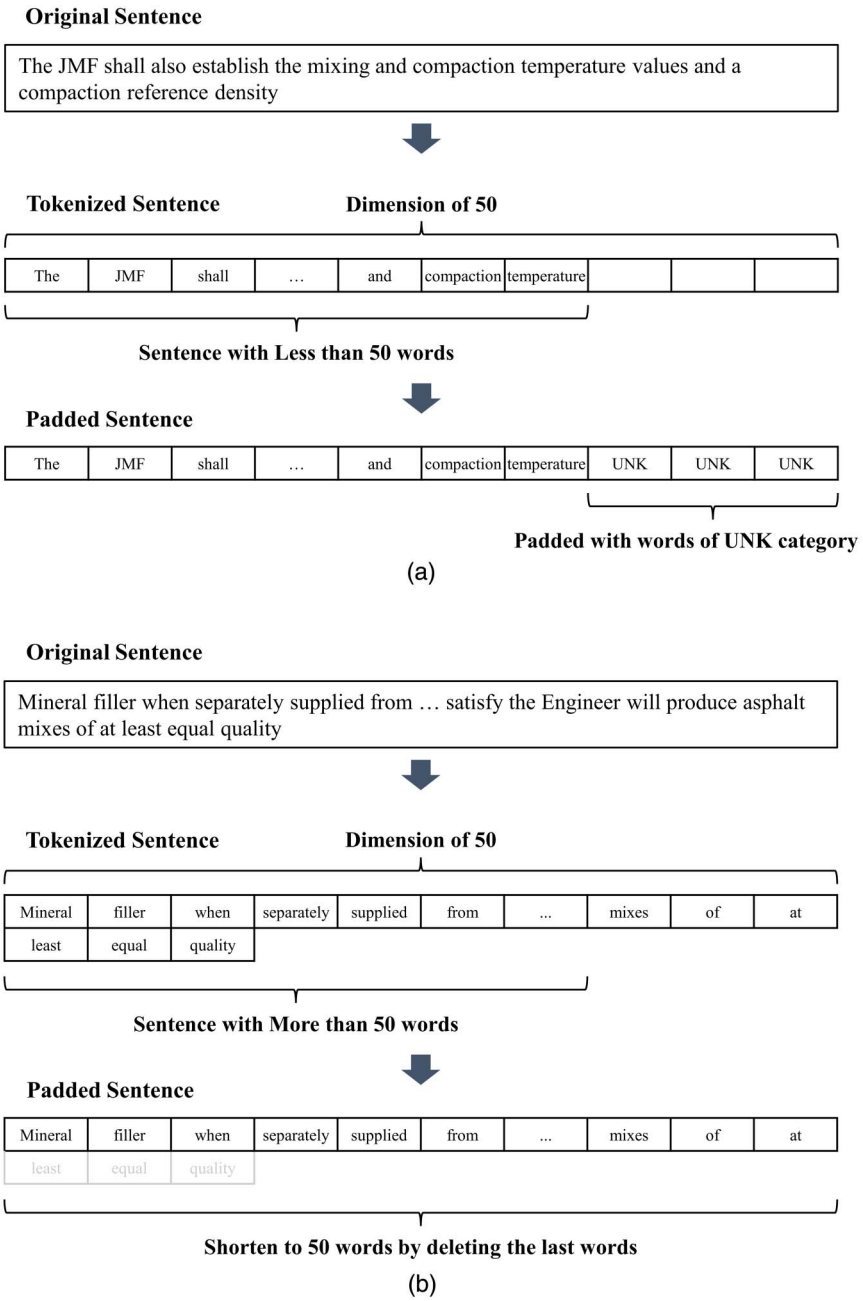


Fig. 10. Sentence padding: (a) sentence less than 50 words; and (b) sentence more than 50 words.

and testing the NER model. They read every sentence and assigned the appropriate category to every word in the sentence (i.e., ORG, ACT, ELM, STD, or REF, and NON for words that did not play a role in understanding the context). Every labeled sentence was cross-checked by each practitioner to ensure that the data were labeled consistently.

Model Input Adjustment

Every sentence was tokenized into a sequence of words and, at the same time, was padded to the length of 50 words. Sentences with less than 50 words were extended with new tokens assigned to an Unknown (UNK) category [Fig. 10(a)], and, conversely, sentences with more than 50 words were shortened by deleting the last words [Fig. 10(b)]. The maximum length of the sentence was set to 50 words according to the experiential knowledge that 50 words would be sufficient to contain a meaningful context in construction specifications. Trial-and-error experiments showed that 50 words or more were too many for a maximum sentence length, as shorter sentences being padded with a large number of UNK tokens.

After adjusting the length of the sentences to 50 words, each word in the sentences was embedded using Word2Vec and mapped into the vector space with a unique numeric vector. The model parameters were customized, including vector size (i.e., the dimension of the embedded word vector), window size (i.e., the considered span of the distribution of words), and epochs (i.e., the number of learning times).

NER Model Training

The Bi-LSTM model used text sentences (i.e., a sequence of words) as the input, and it returned category labels (i.e., a sequence of

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Table 4. Examples of Word2Vec embedding results

Word	Most similar words
AASHTO	ASTM, standard, test
Bituminous	Asphalt, polymer, surfacing, mixtures
Contractor	Engineer, his, responsible
Failure	Stress, event, defects, load
Weather	Conditions, event, protect, wet

labels) as the output. In order to achieve the most appropriate output, the model analyzed the input and extracted the features that consisted of (1) the numerical values from each word vector, (2) the bidirectional order of the words in each sentence, and (3) the bidirectional order of the labels in each sentence. The model derived an identical output vector for each word vector and classified the label based on the output vector. The parameters of the model were configured with the input dimension of 50 words (i.e., the maximum length of a sentence), the output dimension of 100 (i.e., the dimension of the output vector), a dropout of 0.2, a batch size of 32, and epochs of 200 by grid search.

The NER model was trained using 70% of the data (3,261 sentences) and validated using the remaining 30% of the data (1,398 sentences). The training and validation data were partitioned randomly, irrespective of the origin of each sentence (i.e., which document each sentence came from) to develop a robust model by learning a range of text information with different expressions. Furthermore, the model trained 30 different sets of randomly split training and validation data to avoid overtraining on a specific data set.

Experimental Results

Embedding Results of Word2Vec

The research team conducted the whole procedure in CentOS Linux release 7.5 and Python version 3.6.7. The libraries used for the methodology development included gensim-3.6.0, Keras-1.0.5, and TensorFlow-1.11.0.

The Word2Vec model was developed with the following parameters: vector size 200, window size 10, and epochs 200 by grid search. The model seemed to train the representations of the text with satisfactory performance. As shown in Table 4, "ASTM" was most similar to the term "AASHTO" throughout the documents—both terms were used as fundamental standards. The words "bituminous," "failure," and "weather" were synonymous with "asphalt," "stress," and "conditions," respectively; thus, the model seemed to train these features successfully. In addition, the words "contractor" and "engineer," which were examples from the word-embedding section, were identified as being the most similar to each other.

Prediction Results of the NER Model

Fig. 11 provides an example of the NER results in which every word of the original text [Fig. 11(a)] was automatically tagged as belonging to one of the six word categories [Fig. 11(b)]. The sentences were sampled randomly from the validation set. As indicated in the figure, every word was assigned one by one to the relevant category. Although there were some misclassifications (e.g., "total" was assigned to NON, "percent" was assigned to ELM, and "sieve" was assigned to STD) of the 85 identification results, 75 appeared to be consistent with common practical knowledge.

Table 5 is a confusion matrix of the NER classification results, explaining both the original and predicted categories (i.e., ORG, ACT, ELM, STD, REF, and NON). There were 30,109 tokens in the validated set of 1,398 sentences (i.e., 30% of the total labeled data). Although the labeled volume of each category was unbalanced, the classification performance was stable and satisfactory.

Table 6 explains the numerical classification performance of each category. The F1 score is one of the indexes used to measure the accuracy of a model, system, or test (Manning et al. 2008). The F1 score simultaneously considers two well-known performance indexes—precision and recall—by calculating their harmonic means. The developed NER model resulted in an average F1 score of 0.917, indicating that the model was ready for use in real-world applications. In addition, the model's precision and recall were calculated as 0.919 and 0.914, respectively, indicating that the model predicted the category as accurately as the actual extracted category information. The training results using 30 different sets of randomly split training and validation data confirmed the robustness of the model and that it was not overfitted to the specific data set. The results showed an average F1 score of 0.912, with a minimum of 0.900 and a maximum of 0.926 (Table 7).

In detail, the ORG and REF classification results showed the highest accuracy (i.e., 0.945 and 0.946, respectively), despite making up the least volume in the training set. These results likely occurred because the words relating to the two categories comprised only a few terms and were written in an extremely structured

Original Text

The design and quality control of ACHM surface course mix shall be according to Section 404. Design Requirements for Asphalt Concrete Hot Mix Surface Course (1/2inch [12.5 mm]). Fines to asphalt ratio shall be defined as the percent materials passing the No. 200 (0.075 mm) sieve (expressed as a percent of total aggregate weight) divided by the effective asphalt binder content. (a) Mineral aggregate will be measured by the ton (metric ton). Additives for liquid asphalt, when required or permitted, shall meet the requirements of Subsection 702.08.

(a)

NER Results

The$_{[NON]}$ design$_{[ELM]}$ and$_{[NON]}$ quality$_{[ELM]}$ control$_{[ELM]}$ of$_{[NON]}$ ACHM$_{[REF]}$ surface$_{[ELM]}$ course$_{[ELM]}$ mix$_{[ELM]}$ shall$_{[ACT]}$ be$_{[ACT]}$ according$_{[NON]}$ to$_{[NON]}$ Section$_{[REF]}$ 404$_{[REF]}$. Design$_{[ELM]}$ Requirements$_{[ELM]}$ for$_{[NON]}$ Asphalt$_{[ELM]}$ Concrete$_{[ELM]}$ Hot$_{[ELM]}$ Mix$_{[ELM]}$ Surface$_{[ELM]}$ Course$_{[ELM]}$ (1/2inch$_{[STD]}$ [12.5 mm])$_{[STD]}$. Fines$_{[ELM]}$ to$_{[NON]}$ asphalt$_{[ELM]}$ ratio$_{[ELM]}$ shall$_{[ACT]}$ be$_{[ACT]}$ defined$_{[ACT]}$ as$_{[NON]}$ the$_{[NON]}$ percent$_{[ELM]}$ materials$_{[ELM]}$ passing$_{[ACT]}$ the$_{[NON]}$ No.$_{[STD]}$ 200$_{[STD]}$ (0.075$_{[STD]}$ mm)$_{[STD]}$ sieve$_{[STD]}$ (expressed$_{[NON]}$ as$_{[NON]}$ a$_{[NON]}$ percent of$_{[NON]}$ total$_{[NON]}$ aggregate$_{[ELM]}$ weight)$_{[NON]}$ divided$_{[NON]}$ by$_{[NON]}$ the$_{[NON]}$ effective$_{[ELM]}$ asphalt$_{[ELM]}$ binder$_{[ELM]}$ content$_{[ELM]}$. (a)$_{[NON]}$ Mineral$_{[ELM]}$ aggregate$_{[ELM]}$ will$_{[ACT]}$ be$_{[ACT]}$ measured$_{[ACT]}$ by$_{[NON]}$ the$_{[NON]}$ ton$_{[ELM]}$ (metric$_{[ELM]}$ ton)$_{[ELM]}$. Additives$_{[ELM]}$ for$_{[NON]}$ liquid$_{[ELM]}$ asphalt$_{[ELM]}$, when$_{[NON]}$ required$_{[NON]}$ or$_{[NON]}$ permitted$_{[ACT]}$, shall$_{[ACT]}$ meet$_{[ACT]}$ the$_{[NON]}$ requirements$_{[NON]}$ of$_{[NON]}$ Subsection$_{[REF]}$ 702.08$_{[REF]}$.

(b)

Fig. 11. Examples of experimental results: (a) original text; and (b) NER results.

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Table 5. NER confusion matrix

Predicted categories	Actual categories						
	NON	ORG	ACT	ELM	STD	REF	Total
NON	10,360	10	382	527	63	26	11,368
ORG	14	571	0	28	0	2	615
ACT	366	0	4,409	82	15	1	4,873
ELM	694	12	86	9,273	138	22	10,225
STD	62	0	25	135	1,764	13	1,999
REF	10	0	2	36	0	981	1,029
TOTAL	11,506	593	4,904	10,081	1,980	1,045	30,109

Table 6. NER performance

Category	Precision	Recall	F1 score
NON	0.900	0.911	0.906
ORG	0.963	0.928	0.945
ACT	0.899	0.905	0.902
ELM	0.920	0.907	0.913
STD	0.891	0.882	0.887
REF	0.939	0.953	0.946
AVG	0.919	0.914	0.917

Table 7. F1 scores of 30 randomly split data sets

Iteration	F1 score
1	0.907
2	0.900
3	0.904
4	0.914
5	0.913
6	0.911
7	0.904
8	0.910
9	0.904
10	0.905
11	0.914
12	0.919
13	0.926
14	0.912
15	0.914
16	0.913
17	0.901
18	0.917
19	0.918
20	0.918
21	0.919
22	0.921
23	0.910
24	0.907
25	0.920
26	0.924
27	0.908
28	0.911
29	0.916
30	0.914

format within the text. For instance, the ORG category included words such as "Engineer," "Contractor," and "Manager," which were placed near causative verbal phrases such as "should submit to," "must prepare," and "is responsible for." In addition, the REF category included words such as "ASTM C 535," "AASHTO T 245," and "BS EN 12697," which appeared after prepositional phrases, such as "in accordance with," "according to," and "as determined by." However, the model showed relatively less accurate performance for the ACT and STD categories (i.e., 0.902 and 0.887, respectively). The ACT category included various phrases, such as "be based on," "will produce," and "shall be in accordance with," expressed using multiple verbs and prepositions; thus, the category was difficult to differentiate according to different usage purposes. For instance, the word "in" should be assigned to the ACT category in the case of "shall be *in* a good condition," but the NON category is more appropriate in the case of "supply adequate *in* order to execute." Meanwhile, the STD category included terms such as "55%," "175°C," and "2.36 mm," which corresponded closely in terms of format and usage patterns to terms included in the ELM category, such as "60/70 penetrations bitumen," "a 3 m long straight edge," and "MC-70 liquid asphalt." For example, "No. 200 sieve" should have been assigned to the ELM category, but the model incorrectly assigned it to the STD category.

Conclusion

Despite the need for a clear and objective understanding of construction specifications during the bidding stages, it is difficult to review detailed provisions due to limitations of the manual process. The automated models proposed in previous studies have focused on extracting information regarding several types of construction elements from specific clauses of specifications with manually built extraction rules. The current research proposed information extraction frameworks to address the expandability of the automatic review process in two aspects. The authors defined five information types (i.e., the word categories of the NER model) to understand the various types of specification contents via a case study. The authors then developed a machine learning–based NER model to extract the user-needed information automatically. A total of 4,659 sentences sampled from 56 construction specifications were labeled according to the defined information categories. Word2Vec was used to convert the labeled text data into the form of numeric vectors to be input into the NER model. The Bi-LSTM architecture was used for the NER development, and the model successfully assigned every word to an appropriate category with a satisfactory performance of 0.919 precision and 0.914 recall. The construction practitioners who collaborated in the research agreed that the results indicate the model's ability to facilitate the review of construction specifications in practice by identifying the necessary information quickly and accurately.

The results of this research contribute to the construction body of knowledge. The NER model successfully extracted the user-needed information from the specifications (i.e., persons and organizations in charge, activities required, construction and installation items, quality standards and criteria, and relevant references). The construction specifications were able to be understood automatically based on these information types regardless of volume, formats, and organizational structures of clauses. Furthermore, the specifications were able to be reviewed quickly and accurately without any omission of information, and the results were consistent. In addition, the research has the potential to contribute to the construction industry by providing necessary information used for contract risk management during the bidding and planning stages, quality management during the construction stage, and conflict management after project completion. Using this model, contractors could check whether construction requirements on the specification meet the local conditions and are consistent over the entire specification during the bidding stages. They would be able to review the provisions from the specification more efficiently and

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find problematic provisions with different standard criteria, which would assist them in proactively planning for potential troubles. In addition, this research is one of the pioneering efforts in applying an RNN-based information extraction model (i.e., Bi-LSTM) to NER in the construction industry. This approach will provide not only methodological support but also lessons learned for other types of construction document analysis.

Nevertheless, there are potential improvement opportunities. First, because data quality directly impacts the analysis performance, the original PDF should be converted to TXT data as cleanly as possible. In this research, the authors had to manually convert the data due to the functional limitations of the open-source conversion software. Better PDF-to-TXT conversion techniques customized for construction specifications need to be developed to replace the manual process in order to enable larger amounts of data to be processed accurately at a lower cost. Second, the more labeled data the model trains, the better performance the model can achieve. Active learning should be considered in order to reduce the sample size for training while still maintaining a high accuracy level and minimizing the researchers' manual labeling efforts. Third, the advantages of the research findings in terms of cost efficiency and subjectivity need to be validated in a real construction project by comparing them with traditional risk management approaches. Finally, the text analysis approaches proposed in this study can be applied to other research, such as design code verification referenced in the specifications and the review of construction contract documents.

Data Availability Statement

Data generated or analyzed during the study are available from the corresponding author by request. Information about the *Journal*'s data-sharing policy can be found here: http://ascelibrary.org/doi/10.1061/(ASCE)CO.1943-7862.0001263.

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